AI/ML MODEL COMPARISON

California Housing Price Prediction using Regression Algorithms

Organization	MainCrafts Technology	**Domain**	Artificial Intelligence & Machine Learning
Task	Task 2 — Model Comparison	**Notebook**	AI_ML_Task2_Model_Comparison.ipynb

1. Project Overview

This project compares three supervised machine-learning regression algorithms for predicting median house values using the California Housing dataset. The goal is to evaluate prediction errors and goodness of fit on held-out test data and identify the strongest model.

2. Dataset and Tools

- **Dataset:** California Housing dataset provided by scikit-learn.
- **Size:** 20,640 records and 8 input features.
- **Target:** MedHouseVal (median house value).
- **Split:** 80% training and 20% testing data.
- **Tools:** Python, Pandas, NumPy, Matplotlib, Seaborn, and scikit-learn.
- **Preparation:** Input features were standardized using StandardScaler.

3. Methodology

1. Loaded the dataset and inspected its structure, summary statistics, columns, and missing values.
2. Prepared input features and target values, then split the dataset into training and testing sets.
3. Trained Decision Tree Regression, Ridge Regression, and Linear Regression models.
4. Generated test-set predictions and evaluated models using MAE, RMSE, and R^2 score.
5. Created comparison charts and actual-versus-predicted visualizations.

4. Model Evaluation Results

Model	MAE	RMSE	R^2 Score
Decision Tree	0.4267	0.6563	0.6713
Ridge Regression	0.5332	0.7456	0.5758
Linear Regression	0.5332	0.7456	0.5758

Metric interpretation: Lower MAE and RMSE indicate smaller prediction errors; a higher R^2 score indicates that the model explains more variation in the target.

5. Best Model and Conclusion

Decision Tree performed best among the three evaluated models, with the lowest RMSE (0.6563), lowest MAE (0.4267), and highest R^2 score (0.6713). On this test split, it produced more accurate predictions than Ridge Regression and Linear Regression.

Possible future improvements include cross-validation, hyperparameter tuning, checking for overfitting, and evaluating additional regression algorithms.

6. Project Deliverables

- AI_ML_Task2_Model_Comparison.ipynb — notebook with analysis, training, metrics, and plots.
- model_comparison_results.csv — saved model comparison metrics.

- README.md — project overview, methodology, results, and conclusion.
- MainCrafts_AIML_Task2_Report.pdf — this report.

Note: The California Housing target is expressed in units of $100,000. The reported MAE and RMSE are in dataset target units, not directly in dollars. Results depend on the current data split and model settings.